Overcoming Data Scarcity for Event-Based Pupil Tracking with Synthetic and Unlabeled Data

AARON TOGNOLI, Politecnico di Milano, Italy
ANDREA SIMPSI, Politecnico di Milano, Italy
ANDREA ASPESI, EssilorLuxottica, Italy and Politecnico di Milano, Italy
MARCO CANNICI, Politecnico di Milano, Italy
LUCA MERIGO, EssilorLuxottica, Italy
MATTEO MATTEUCCI, Politecnico di Milano, Italy
SIMONE MENTASTI, Politecnico di Milano, Italy

Smart eyewear is emerging as an always-on platform capable of perceiving the environment and inferring intent through eye movements, making pupil tracking essential for personalized interaction. However, reliable tracking on wearable hardware remains challenging due to strict power limits and scarce annotated data. Event-based cameras offer a low-power, microsecond-latency solution, but labeled recordings still remain limited. We address this issue with a training framework that combines limited annotated real data with synthetic events and unlabeled real recordings, learning event-based pupil trackers with strong real-world generalization. We pair the U2Eyes tool with the v2e event camera simulator to generate realistic event streams, showing that networks trained on these events exhibit smaller sim-to-real gaps than networks trained on synthetic images. Moreover, our training procedure further bridges this gap, enabling our models to outperform networks trained exclusively on real data across all benchmarks, advancing toward more robust event-based eye tracking on wearable platforms.

CCS Concepts: • **Computing methodologies** → **Interest point and salient region detections**; • **Human-centered computing** → *Interaction techniques*.

Additional Key Words and Phrases: Eye Tracking, Neural Network, Event Camera, Synthetic Data, Smart Eyewear

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1 Introduction

Smart eyewear devices are emerging as always-on companions capable of perceiving the user's surroundings and interpreting gaze behavior [Palermo et al. 2025]. Equipped with embedded eye-tracking algorithms, these devices continuously estimate the pupil's position and dynamics to estimate where the user is looking and provide valuable cues about intent [Belardinelli 2024],

Authors' Contact Information: Aaron Tognoli, DEIB, Politecnico di Milano, Milan, Italy, aaron.tognoli@polimi.it; Andrea Simpsi, DEIB, Politecnico di Milano, Milan, Italy, andrea.simpsi@polimi.it; Andrea Aspesi, EssilorLuxottica, Milan, Italy, DEIB and Politecnico di Milano, Milan, Italy, andrea.aspesi@luxottica.com; Marco Cannici, DEIB, Politecnico di Milano, Milan, Italy, marco.cannici@polimi.it; Luca Merigo, EssilorLuxottica, Milan, Italy, luca.merigo@luxottica.com; Matteo Matteucci, DEIB, Politecnico di Milano, Milan, Italy, matteo.matteucci@polimi.it; Simone Mentasti, DEIB, Politecnico di Milano, Milano, Italy, simone.mentasti@polimi.it.

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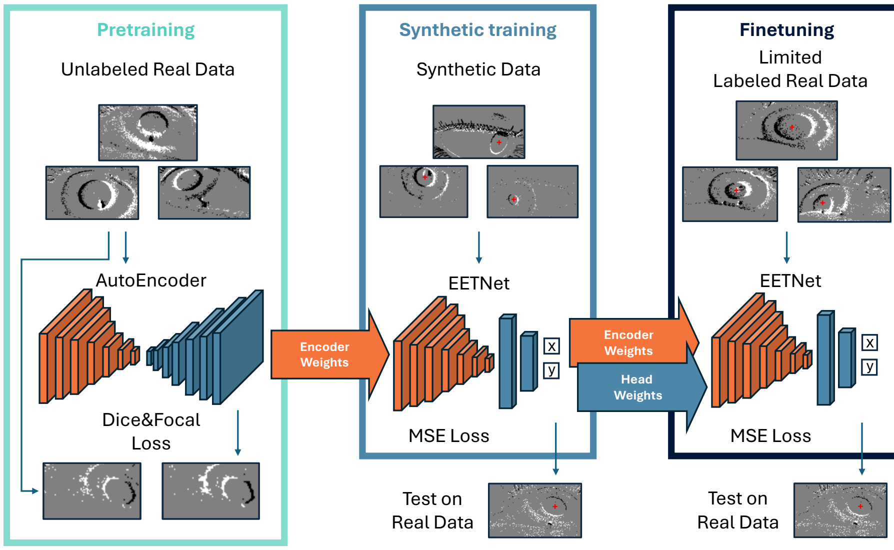


Fig. 1. Overview of the proposed method. The pretraining enables the use of unlabeled data, which, when combined with training on synthetic data and finetuning on real data, enhances generalization and improves model robustness.

cognitive load and health [Ashraf et al. 2018; Loke et al. 2025]. Such capabilities are central to next-generation head-mounted devices that aim to share the user's perceptual scene and deliver adaptive, context-aware interactions [Palermo et al. 2025], making low-latency, high-precision pupil tracking essential to next-generation smart eyewear. Achieving such capability on eyewear is, however, far from trivial. Smart glasses impose strict limits on power, form factor, and computation, leaving little room for conventional camera pipelines or large neural networks [Lin 2025]. While a variety of geometric and learning-based eye-tracking methods exist [de Lope and Graña 2022; Illahi et al. 2022; Lu et al. 2020; Ou et al. 2021; Sun et al. 2021], they typically depend on traditional dense image processing [Moreno-Arjonilla et al. 2024], which is power-intensive and limited by a fundamental bandwidth–latency trade-off: increasing the frame rate to capture rapid eye motion proportionally raises data throughput and computation, hindering real-time operation on wearable devices. Moreover, training deep models that remain reliable in real-world conditions poses an additional challenge, as collecting precisely annotated eye images on head-mounted devices is labor-intensive and time-consuming, and even minor variations in subjects, illumination, or calibration can substantially degrade performance at deployment. As a result, current neural trackers often struggle to generalize across users, limiting their practical adoption on embedded platforms.

To address these limitations, recent work has explored event-based cameras as an alternative sensing modality for eye tracking [Iddrisu et al. 2024a]. Inspired by biological vision, these sensors capture asynchronous per-pixel brightness changes rather than full image frames, producing inherently sparse, motion-sensitive, and energy-efficient outputs with microsecond latency and high dynamic range [Gallego et al. 2020]. These characteristics make event cameras especially well-suited for near-eye settings, where rapid pupil motion imposes demanding temporal requirements on the sensing pipeline. By transmitting only brightness changes, event cameras bypass the bandwidth–latency trade-off, enabling low-latency, efficient tracking even during rapid eye motion. Recent works have confirmed this potential, showing that event-based approaches can achieve high-speed and robust pupil or gaze tracking across a range of designs, from convolutional and spiking neural models to hybrid frame–event pipelines [Aspesi et al. 2025; Jiang et al. 2024; Stoffregen et al. 2022], making them particularly attractive for wearable deployment.

Despite these promising results, however, event-based eye tracking, especially neural-based, remains limited in robustness, as publicly available datasets contain only few recordings with coarse annotations, far fewer than their image-based counterparts, typically derived from on-screen stimulus positions [Angelopoulos et al. 2021] or provided at low temporal resolution [Zhao et al. 2023]. This scarcity largely stems from the novelty of event-based cameras in this domain and the difficulty of annotating their sparse, high-frequency outputs reliably. Simulation represents a promising strategy to mitigate data scarcity by enabling large-scale generation of accurately labeled event data under diverse conditions [Planamente et al. 2021]. However, event-based eye tracking has yet to systematically exploit such simulation pipelines or assess their impact on real-world transfer.

To advance event-based pupil tracking, we introduce a new training framework that leverages synthetic event data and unlabeled real recordings to achieve robust real-world generalization. Specifically, we pair the U2Eyes [Porta et al. 2019] renderer with the v2e [Hu et al. 2021] event simulator to produce realistic event streams under realistic motion and illumination, providing the first systematic analysis of fully simulated event data for eye tracking. Using this pipeline, we first train pupil-tracking networks entirely in simulation and demonstrate stronger zero-shot transfer to real event data than analogous models trained on synthetic images and tested on real frames. Finally, we propose a simple yet effective three-stage training pipeline that combines label-free pretraining on unlabeled real events, supervised learning on synthetic data, and fine-tuning on limited real annotations. This approach, depicted in Figure 1, further reduces the residual domain gap, consistently improving real-world transfer without requiring additional manual annotations.

Through extensive experiments on public event-based eye-tracking datasets, EV-Eye [Zhao et al. 2023], 3ET+ [Wang et al. 2024a], INI30 [Bonazzi et al. 2024], and the dataset by Angelopoulos et al. [Angelopoulos et al. 2021], we demonstrate consistent performance improvements across all benchmarks. Our results show that event-based representations exhibit a smaller sim-to-real gap than image-based counterparts, and that our proposed training pipeline outperforms previous methods trained solely on real event data.

In summary, our main contributions are as follows:

- We combine the U2Eyes software with the v2e simulator to generate synthetic event data and extend the U2Eyes to generate accurate pupil center annotations for training
- We demonstrate that networks trained on synthetic events achieve a smaller sim-to real gap than models trained on simulated images.
- We develop a three-stage training framework that integrates label-free pretraining on unlabeled real events with synthetic and limited real supervision, which improves real-world transfer without requiring additional manual annotation.
- We conduct a comprehensive evaluation on four public event-based datasets, showing consistent accuracy improvements and confirming that simulated event data provide an effective foundation for future event-based eye tracking solutions.

Societal Impact Statement. By reducing dependence on newly collected human data, our simulation-based framework mitigates privacy concerns. More broadly, it provides a generalizable procedure to advance event-based pupil regressors, enabling the improvement of existing approaches and fostering progress in this field.

2 Related Works

Event-based Eye Tracking. A number of studies have explored event-driven pipelines for pupil localization and gaze estimation [Iddrisu et al. 2024a], motivated by the microsecond-level latency and milliwatt power consumption of event cameras. Early implementations focused on geometric approaches designed to minimize latency and energy usage. Angelopoulos et al. [Angelopoulos et al. 2021] combined frame and event-based sensing for updates beyond 10,000 Hz, while Kagemoto et al. [Kagemoto and Takemura 2023] and Stoffregen et al. [Stoffregen et al. 2022] achieved low-latency glint and pupil localization via active and coded illumination schemes.

Deep learning has also been widely explored to enhance accuracy and robustness. Several approaches reconstruct pseudo-images from temporally binned events and process them using convolutional networks. Among these, Iddrisu et al. [Iddrisu et al. 2024b] employed standard CNN architectures such as GR-YOLO [Ryan et al. 2021] and YOLOv8 [Jocher et al. 2023] to assess the suitability of conventional models, while Aspesi et al. [Aspesi et al. 2025] proposed EETnet, a compact CNN optimized for real-time inference on MAX78000 embedded hardware. In contrast, other methods [Feng et al. 2022; Li et al. 2023; Zhao et al. 2023] proposed lightweight U-Net–based segmentation networks for pupil segmentation, exploiting MobileNet-like [Howard et al. 2017] modules and ROI mechanisms to reduce inference load. To better capture temporal dependencies, more recent methods integrate temporal reasoning into their architectures. Pei et al. [Pei et al. 2024] utilized separable spatiotemporal convolutions, Chen et al. [Chen et al. 2023] exploited convolutional LSTMs in the 3ET framework, and subsequent studies adopted state-space models for continuous-time event processing [Wang et al. 2024b; Zubic and Scaramuzza 2025]. More recently, spiking neural networks have been explored as energy-efficient alternatives for event-based eye tracking. Bonazzi et al. [Bonazzi et al. 2024] achieved real-time pupil tracking on neuromorphic hardware using a lightweight SNN architecture, while Jiang et al. [Jiang et al. 2024] employed a SEW-ResNet backbone with temporal coding mechanisms to more effectively capture the spatial and dynamic features of event streams. While these learning-based methods have demonstrated strong performance, their effectiveness is often constrained by the limited availability and diversity of training data, which hinders generalization to real-world scenarios. In contrast, our approach leverages both unannotated real event streams and synthetically generated data, enhancing robustness and adaptability.

Near-Eye Synthetic Image Rendering. Motivated by the need for dense labels and controllable conditions, synthetic renderers for image-based eye tracking evolved from geometry-based pipelines to physically-based and scalable simulators. Early work by Sugano et al. [Sugano et al. 2014] used multi-view reconstruction and reprojection to synthesize gaze poses and increase gaze coverage, while Świrski and Dodgson [Świrski and Dodgson 2014] and Wood et al. [Wood et al. 2015] leveraged anatomically inspired models and HDR lighting to render photorealistic images with exact ground truth. Later approaches employed real-time rendering in UnityEyes [Wood et al. 2016] to enable large-scale data generation, further extended by U2Eyes [Porta et al. 2019] with binocular synthesis and improved realism. To enhance realism, SimGAN [Shrivastava et al. 2017] leveraged adversarial training with unlabeled real images to improve realism, whereas recent NeRF-based approaches, including GazeNeRF [Ruzzi et al. 2023] and NeRF-Gaze [Yin et al. 2024], achieve photorealistic synthesis via neural radiance fields at increased computational cost. Despite significant progress in synthetic and neural rendering, event-based eye-tracking research remains limited by the scarcity of annotated real datasets. Current datasets, such as Angelopoulos et al [Angelopoulos et al. 2021], 3ET [Chen et al. 2023], 3ET+ [Wang et al. 2024a], INI-30 [Bonazzi et al. 2024], EV-Eye [Zhao et al. 2023], and RGBE-Gaze [Zhao et al. 2024], include only a few dozen participants and sparse annotations, with the largest to date containing just 66 subjects, thereby constraining the generalization capacity of learned models. To address this limitation, we adapt the

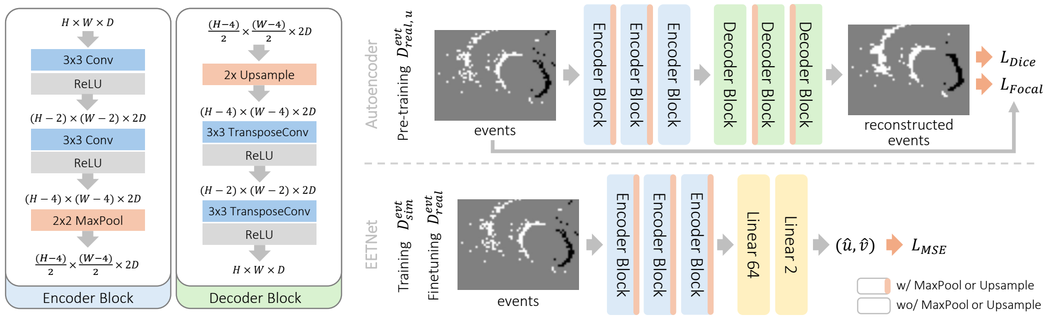


Fig. 2. Overview of the architectures used in this work. The top part illustrates the autoencoder, whose encoder is derived from EETNet. The autoencoder is trained on unlabeled real samples $\mathcal{D}^{\text{evt}}_{\text{real,u}}$, and the learned encoder weights are then used to initialize EETNet (shown below), which is subsequently trained on synthetic $\mathcal{D}^{\text{evt}}_{\text{sim}}$ and real $\mathcal{D}^{\text{evt}}_{\text{real}}$ samples. The detailed structure of the encoder and decoder blocks is shown on the left.

U2Eyes simulator [Porta et al. 2019] to generate synthetic event streams using the v2e [Hu et al. 2021] event simulation framework from high-frequency near-eye renderings, enabling large-scale, fully annotated data for event-driven eye tracking.

3 Method

Existing event-based camera datasets exhibit two key limitations: manual annotation introduces inherent labeling noise, and annotations are available only at sparse temporal intervals, far below the microsecond resolution of the recorded events. Simulation addresses these issues by providing dense, perfectly aligned labels under controlled geometry and motion. Nevertheless, synthetic events differ from real recordings in noise, texture, and dynamics, resulting in a *domain shift* that can degrade performance on real-world data. Our method mitigates this gap through a multi-stage pipeline that leverages simulation to train event-based pupil trackers and later adapt them to real-world data. Specifically, we first perform supervised learning on synthetic events and subsequently fine-tune the model on limited annotated real recordings. Optionally, when unlabeled real data is available, we demonstrate that an unsupervised label-free pretraining stage further improves transfer and generalization on real event streams. In the following, we formalize the learning problem under simulation, describe the generation of realistic event streams in Section 3.1, and present our complete training framework in Section 3.2.

Problem Formulation: Let us formally define the problem of transferring a network trained on simulated data to real sensor data. We denote by $\mathcal{X}^{\text{img}}$ and $\mathcal{X}^{\text{evt}}$ the input spaces corresponding to the image and event modalities, respectively. In the image modality, each sample $\mathbf{x} \in \mathcal{X}^{\text{img}} = \mathbb{R}^{H \times W}$ represents an intensity frame captured at a discrete time instant, where H and W denote the image height and width. In contrast, an event-based sensor produces a stream of asynchronous events $\mathcal{E} = \{e_i = (u_i, v_i, t_i, p_i)\}_{i=1}^{N} \in \mathcal{X}^{\text{evt}}$, where $(u_i, v_i) \in [0, W) \times [0, H)$ denotes the pixel location, t_i the timestamp, and $p_i \in \{+1, -1\}$ the polarity of the brightness change. The event input space $\mathcal{X}^{\text{evt}}$, therefore, corresponds to the set of spatio-temporal event sequences observed over a fixed time window.

Each modality can exist in two domains: a simulated domain, obtained through rendering or simulation, and a real domain, acquired from physical sensors. We denote these domains as $\mathcal{D}^{\text{img}}_{\text{sim}}$, $\mathcal{D}^{\text{img}}_{\text{real}}$, $\mathcal{D}^{\text{evt}}_{\text{sim}}$, $\mathcal{D}^{\text{evt}}_{\text{real}}$, corresponding to simulated and real domains for both image and event modalities. In this work, we focus primarily on the event modality, where the goal is to train a

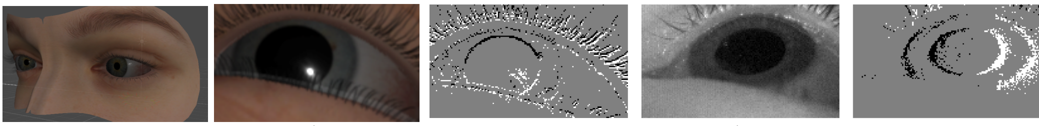


Fig. 3. Comparison between data generated from the renderer and event camera simulator and that recorded from real camera sensors in Angelopoulos et al. From left to right: (a) 3D eye model, (b) simulated image ($\mathcal{D}_{\text{sim}}^{\text{img}}$), (c) simulated events ($\mathcal{D}_{\text{sim}}^{\text{evt}}$), (d) real image ($\mathcal{D}_{\text{real}}^{\text{img}}$), and (e) real events ($\mathcal{D}_{\text{real}}^{\text{evt}}$).

model on synthetic events sampled from $\mathcal{D}_{\text{sim}}^{\text{evt}}$ and deploy it on real events drawn from $\mathcal{D}_{\text{real}}^{\text{evt}}$. We explore multiple training regimes: (i) training solely on synthetic events from $\mathcal{D}_{\text{sim}}^{\text{evt}} = \{(x_i, y_i)\}$ with $x_i \in \mathcal{X}^{\text{evt}}$ and pupil center annotations $y_i \in \mathbb{R}^2$ (*sim-only*); (ii) training solely on *limited* real events from $\mathcal{D}_{\text{real}}^{\text{evt}}$ (*real-only*); (iii) joint training on both domains $\mathcal{D}_{\text{sim}}^{\text{evt}} + \mathcal{D}_{\text{real}}^{\text{evt}}$ (*joint*); and (iv) a two-stage setup, where the model is first trained on synthetic data and subsequently fine-tuned on real events (*train + finetune*). To further investigate generalization, we also consider two additional variants: (v) a *three-stage* configuration, where unlabeled real samples $\mathcal{D}_{\text{real,u}}^{\text{evt}} = \{x_j\}$ are additionally incorporated to better align the model with the real data distribution; and (vi) a *zero-shot* configuration, where the network trained on $\mathcal{D}_{\text{sim}}^{\text{evt}}$ is directly evaluated on $\mathcal{D}_{\text{real}}^{\text{evt}}$ without adaptation.

3.1 Rendering Realistic Eye Tracking Events

We construct simulated events $\mathcal{D}_{\text{sim}}^{\text{evt}}$ by first rendering high-frequency eye images $\mathcal{D}_{\text{sim}}^{\text{img}}$ using an extended version of the U2Eyes [Porta et al. 2019] simulator, and then converting those frames into event streams with the v2e [Hu et al. 2021] event camera simulator. Based on UnityEyes [Wood et al. 2016], U2Eyes provides a 3D model of the human eye region derived from high-resolution face scans, featuring anatomically inspired eyelid animation and physically based eyeball materials. It supports variable face models and gaze direction, multiple skin tones and iris textures, and outputs 2D and 3D landmarks for eyelids, iris, and pupil.

3.1.1 Accurate eye rendering. We extend the U2Eyes simulator [Porta et al. 2019] to generate geometrically accurate pupil annotations by explicitly modeling corneal refraction. In the original implementation, the pupil center is approximated by reprojecting onto the camera frame a point on the external corneal surface along the pupil–bulb axis, which leads to significant localization errors at oblique viewing angles common in near-eye tracking setups. We address this limitation by computing the refracted pupil center as perceived by the camera. In practice, pupil annotations are obtained by rendering a binary mask of the refracted pupil using Unity's refraction model. This is achieved by temporarily replacing the eye bulb material with a uniform black texture and assigning a distinct color to the 3D pupil ellipse, while preserving the cornea's refractive and reflective properties. The pupil center is then computed as the centroid of the rendered mask, providing pixel-accurate annotations for the gaze and camera configuration. Additionally, we enhance realism by supporting infrared light sources and recording first-surface corneal reflections in the camera frame.

3.1.2 Event camera simulation. Given a few representative samples from $\mathcal{D}_{\text{real}}^{\text{evt}}$, we generate synthetic data by manually adjusting the virtual camera intrinsics and pose to approximately match the real dataset viewpoint. This calibration requires only coarse alignment of the virtual and real camera views. We then render synthetic eye images along a continuous *S-shaped* gaze trajectory spanning horizontal and vertical directions, producing high-frequency sequences at 1000 Hz that cover the full range of eye motion. To increase diversity, each sequence is repeated multiple times while varying the pupil diameter. The rendered image sequences are converted into event streams using the v2e event simulator [Hu et al. 2021], which models an event-based sensor by generating asynchronous events and detecting per-pixel brightness changes over time. This process transforms the rendered images in $\mathcal{D}_{\text{sim}}^{\text{img}}$ into temporally aligned event streams in the event domain $\mathcal{D}_{\text{sim}}^{\text{evt}}$. The simulator exposes several parameters controlling event generation, including the positive and negative contrast thresholds (Θ_+, Θ_-), which specify the minimum logarithmic intensity change required to trigger events of each polarity, and the threshold variability standard deviation (σ_Θ). Since v2e presets are tuned for real video and tend to overestimate event rates on rendered images, we manually adjust these parameters to better match the real event camera. Figure 3 compares the rendering and simulation stages.

3.2 Learning robust trackers through supervised and unsupervised training

Event-based pupil tracking can be approached with a variety of network architectures, ranging from complex backbones [Wang et al. 2024b; Zubic and Scaramuzza 2025] to lightweight spiking [Bonazzi et al. 2024; Jiang et al. 2024] or recurrent models [Chen et al. 2023]. In this work, we focus on compact, embeddable networks suitable for deployment on wearable platforms with strict power and latency constraints. Specifically, we adopt EETNet [Aspesi et al. 2025], a convolutional neural network designed to run efficiently on the MAX78000 [Inc. 2020] low-power CNN accelerator. While our analysis is agnostic to the network architecture, the simplicity and low computational cost of EETNet make it an ideal testbed for evaluating whether the proposed training procedure can improve the performance of models suitable for on-device deployment in smart eyewear. In the following, we outline the training pipeline and model architecture used in this work. Section 3.2.1 details the selected network and its training on synthetic and real recordings, while Section 3.2.2 introduces the unsupervised pretraining stage exploiting unlabeled real events.

3.2.1 Supervised training on synthetic and real events. EETNet follows a minimalist CNN architecture to fit hardware constraints of the MAX78000 microprocessor. The backbone, reported in Figure 2, consists of three convolutional blocks, each composed of two convolutional layers with ReLU activations and a max-pooling layer. The feature maps are then passed to a regression head composed of two fully connected layers. The first processes the flattened convolutional output, while the second predicts the pupil center coordinates $(\hat{u}, \hat{v})$ in the input frame.

Event cameras produce a sparse, asynchronous stream of brightness changes rather than conventional image frames. Following the strategy in EETNet [Aspesi et al. 2025], we convert these asynchronous events into dense event frames by integrating them over short temporal windows. Specifically, events are aggregated at each pixel summing their polarity. Then each pixel value is clipped in the range[-1,1], forming a three-valued image where each pixel takes values in $\{-1, +1, 0\}$ corresponding to negative, positive, or no changes in brightness, respectively. Each accumulated frame is cropped around the eye region and resized to the network's input resolution. Standard data augmentation, including horizontal and vertical flips and random shifts, is applied to improve robustness to head pose and inter-subject variability. While the original EETNet is trained exclusively on real event camera data using a mean squared error (MSE) loss on the pupil center coordinates, in this work we explore multiple training regimes leveraging the simulated and real domains introduced in Section 3. As analyzed in detail in Section 4.3, we found that mixing simulated and real events yields the best performance in our case. Supervised training on simulated data leverages abundant from $\mathcal{D}_{\text{sim}}^{\text{evt}}$ to best exploit pixel-perfect annotations, while fine-tuning on

$\mathcal{D}^{\text{evt}}_{\text{real}}$ adapts the model to real sensor statistics and residual domain differences. In contrast, training solely on limited real data tends to induce overfitting, preventing robust real-world deployment.

3.2.2 Unsupervised pretraining on real events. To reduce the domain gap between synthetic and real data, we introduce an unsupervised pretraining stage that leverages unlabeled real events $\mathcal{D}^{\text{evt}}_{\text{real,u}}$. Since such recordings can be collected effortlessly and are often available in large quantities, they provide a valuable resource for adapting the model to real sensor statistics before supervised training. This stage serves as a label-free initialization that complements our synthetic pretraining and improves generalization on real event streams. To implement this stage, we adopt a simple yet effective strategy based on event-frame reconstruction. Specifically, we train an autoencoder that uses the convolutional encoder of the EETNet pupil regressor, enabling its weights to be initialized with features already adapted to real event statistics. The autoencoder, illustrated in Figure 2, reuses the convolutional feature extractor of EETNet up to the third block as the encoder, while a symmetric decoder composed of transposed convolutions mirrors its structure to reconstruct the input event frame. The model is trained to reproduce each input frame from its latent representation, encouraging the encoder to capture the spatial structure and polarity distribution characteristic of real event recordings. Inspired by semantic segmentation networks, we frame event reconstruction as a three-class segmentation problem over the $\{-1, 0, +1\}$ event polarities in our event representation. To this end, we combine Dice [Fidon et al. 2018] and Focal [Lin et al. 2018] losses, naturally addressing the strong class imbalance dominated by the background (0) class. This formulation proves more effective than a traditional autoencoder trained with mean squared error (MSE), producing sharper and more spatially consistent reconstructions. A visual example of the reconstructed event frames is shown in Figure 2.

In particular, the Dice loss evaluates the overlap between predicted and ground-truth event polarities across all classes:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{C} \sum_{c=0}^{C-1} \frac{2 \sum_{n=1}^{N} p_{n,c}\, \hat{p}_{n,c}}{\sum_{n=1}^{N} \left(p_{n,c} + \hat{p}_{n,c}\right)}, \tag{1}$$

where $C = 3$ is the number of event polarity classes and N the number of pixels in the batch. Here, $p_{n,c}$ and $\hat{p}_{n,c}$ denote the ground-truth and predicted probability for class, i.e., polarity, c at pixel n, respectively. To further emphasize minority classes and penalize misclassified polarities, we complement it with a Categorical Focal loss:

$$\mathcal{L}_{\text{Focal}} = \frac{1}{N} \sum_{n=1}^{N} \left[-\alpha_c (1 - p_{n,c})^{\gamma} \log(p_{n,c})\right], \tag{2}$$

where $p_{n,c}$ is the predicted probability for class c for the sample n, in our case the event polarity of each pixel, α_c is a per-class weighting coefficient, and γ controls the focusing strength. The final reconstruction objective combines the two components as

$$\mathcal{L}_{\text{Total}} = \lambda_{\text{D}} \mathcal{L}_{\text{Dice}} + \lambda_{\text{F}} \mathcal{L}_{\text{Focal}}, \tag{3}$$

where λ_{D} and λ_{F} control the relative contribution of each term.

The pretrained encoder weights are then used to initialize the pupil-tracking feature extraction backbone, providing a feature representation better aligned with real event statistics and improving stability during supervised training.

4 Experimental Validation

4.1 Implementation Details

All models were implemented in PyTorch and trained on a single NVIDIA RTX A6000 GPU. Event streams were accumulated into event frames as described in Section 3.2.1, producing three-valued images used for training. Performance is evaluated using Mean Pupil Distance (Mean Euclidean Distance between predicted and ground-truth pupil centers) and Mean Absolute Error (Mean Manhattan Distance).

Supervised training on simulated or real events is performed for 200 epochs using a mean squared error (MSE) loss on pupil-center coordinates. Unless otherwise stated, each frame is cropped to a 157×90 region centered on the labeled pupil position. To improve generalization, we apply data augmentation including horizontal, vertical, and bidirectional flips, random translations of up to ± 30 pixels, and a 25% zoom centered on the eye. Augmented frames are then downsampled to 78×45 pixels by subsampling odd rows and columns. Dataset-specific details are provided in Section 4.2. All models are trained using the Adam optimizer [Kingma 2014] with parameters $(\beta_1, \beta_2) = (0.9, 0.999)$, batch size 100, and initial learning rate 1×10^{-3}. A Reduce-on-Plateau scheduler halves the learning rate after five epochs without improvement, down to a minimum of 1×10^{-5}.

The *autoencoder* described in Section 3.2.2 is trained on unlabeled real event frames using the same optimizer, batch size, and learning rate. No data augmentation is applied. The reconstruction objective combines Dice and Focal losses with equal weights $\lambda_{\mathrm{D}} = \lambda_{\mathrm{F}} = 1$, class weights $\alpha = [5, 1, 5]$ for the $\{-1, 0, +1\}$ event classes, and focusing parameter $\gamma = 2$. Training lasts 200 epochs, after which only the encoder weights are retained to initialize the pupil-tracking network.

Fine-tuning on real data retrains models pre-optimized on synthetic events for 100 epochs. A cosine learning rate schedule decays from 1×10^{-3} to 1×10^{-5} over the first 80 epochs, followed by 20 epochs at a fixed rate of 1×10^{-5}. The same augmentations as in simulation training are applied, while all other hyperparameters remain unchanged.

4.2 Datasets

4.2.1 EV-Eye. The EV-Eye dataset [Zhao et al. 2023] includes recordings from 48 participants (ages 21–35) and combines near-eye grayscale images with asynchronous event streams captured using two DAVIS 346 cameras (346×240 pixels), along with gaze references from a Tobii Pro Glasses 3 eye tracker. Ground truth includes Points of Gaze, pupil diameter, and approximately 9,000 manually annotated ellipse-shaped pupil labels. In our experiments, we use the latter as pupil-center ground truth, computed as the center of each ellipse. The dataset is split by participant index into training (subjects $1 \rightarrow 38$), validation ($39 \rightarrow 43$), and test ($44 \rightarrow 48$) sets. Training annotations are used to compute a per-user centroid, around which each image is cropped to a 157×90 region of interest. Corresponding event streams are accumulated to generate event-based frames aligned with the cropped region.

4.2.2 INI-30. The INI-30 dataset [Bonazzi et al. 2024] contains recordings from 30 participants performing free-viewing tasks with natural head motion using a custom glasses-mounted setup with two DVXplorer cameras (640×480 pixels). Pupil center annotations are provided directly in the camera frame, offering a realistic "in-the-wild" setting for event-based gaze estimation. Recording durations range from 14.6 s to 193.8 s per subject, with 475 to 1,848 annotated frames per sequence. We preprocess the events in two ways. In the first, following the original protocol, the rectangular sensor resolution is converted to a square 512×512 array by padding the y-axis by 16 pixels on each side and discarding unlabeled and with sparse event activity regions along the x-axis ($x < 96$, $x > 608$). Frames and labels are then sum-pooled to a final resolution of 64×64, yielding the *INI-30* 64×64 variant. In the second variant, frames are processed at the original resolution by cropping a 157×90 region of interest, compatible with EETnet. This is obtained by computing the centroid of the training labels and extracting a centered crop, following the same procedure as Section 4.2.1.The dataset is split by participant index into training (subjects $1 \rightarrow 20$), validation (subjects $21 \rightarrow 25$), and test (subjects $26 \rightarrow 30$) sets.

4.2.3 Angelopoulos et. al. The Angelopoulos et al. dataset [Angelopoulos et al. 2021], referred to as *EB-NEG* in the following, provides near-eye grayscale frames and asynchronous event streams for high-temporal-resolution event-based near-eye gaze estimation. It includes recordings from 24 participants with head fixed on a headrest, performing controlled eye-movement tasks. Grayscale images are captured using two DAVIS 346b event cameras (346×260 pixels) and include corresponding event streams and ground-truth stimulus display coordinates from a 1920×1080 monitor covering approximately $(64^{\circ} \times 96^{\circ})$ of visual angle. The recordings include both random saccade and smooth pursuit tasks.Since our focus is pupil center regression, we use the pupil annotations introduced by Simpsi et al. [Simpsi et al. 2024], which provide pupil locations directly in the camera frame. The dataset is split by participant index into training (subjects 10, 18, 20), validation (subject 19), and test (subjects 5, 15) sets. Event preprocessing follows Section 4.2.1: pupil annotations are used to compute a per-user centroid, and a 157×90 region of interest is cropped for both frames and events.

4.2.4 3ET+. The 3ET+ dataset [Wang et al. 2024a] includes near-eye event streams recorded with a DVXplorer Mini camera from 13 participants, each contributing 2–6 sessions. Subjects performed five activity types: random movements, saccades, text reading, smooth pursuits, and blinks. Ground truth annotations are provided at 100 Hz and include a binary blink label and manually annotated pupil-center coordinates. We adopt the train–test split provided by the authors and follow the event pre-processing procedure described in Section 4.2.1. Additionally,as in Section 4.2.2, to ensure comparability with prior work, we evaluate our method using the original preprocessing protocol, where frames and labels are downsampled by a factor of 8, reducing the resolution from 640×480 to 60×80.

4.3 Results

4.3.1 Domain Shift Analysis. We begin our experimental validation by analyzing the impact of the synthetic-to-real domain shift in a purely supervised setting, without applying pretraining or fine-tuning. Specifically, we train the network on one domain and directly evaluate it on real test data from the corresponding dataset. We consider two types of domain shifts. The first concerns the image modality and quantifies the performance degradation observed when a model trained on synthetic images $\mathcal{D}_{\text{sim}}^{\text{img}}$ is evaluated on real eye images $\mathcal{D}_{\text{real}}^{\text{img}}$. Analogously, the second focuses on the event modality and measures the performance of a network trained on simulated event streams $\mathcal{D}_{\text{sim}}^{\text{evt}}$ and tested on real event data $\mathcal{D}_{\text{real}}^{\text{evt}}$. To contextualize these results, referred to as *sim-only* in Section 3, we compare them with *real-only* reference performance obtained when the network is both trained and tested on the same real domains, namely $\mathcal{D}_{\text{real}}^{\text{img}}$ and $\mathcal{D}_{\text{real}}^{\text{evt}}$. These configurations provide references that represent the performance achievable when training and testing share the same distribution, allowing us to quantify the degradation introduced by the synthetic-to-real shift.

In Table 1, we observe that models trained on simulated events exhibit a markedly smaller domain gap compared to their image-based counterparts. When trained on synthetic data, the event-based network maintains performance much closer to that of models trained on real events, while image-based models experience a substantial degradation when transferred to real eye images. We attribute this reduced gap to the intrinsic properties of event data: events encode temporal brightness changes and motion edges, which are largely invariant to illumination, texture, and texture appearance. In contrast, image-based models tend to overfit to texture and shading cues that are difficult to reproduce faithfully in simulation, resulting in weaker transfer to real sensors. Interestingly, for the Angelopoulos et al., EB-NEG, dataset, models trained solely on simulated

Table 1. Domain shift analysis on the EV-Eye and the Angelopoulos et al., here shortened as EB-NEG, datasets. We report mean absolute error (MAE, in pixels) and mean pupil distance (MPD, in pixels) for different training and testing domain configurations. **Bold** values denote the best result for each dataset and metric. We indicate with Δ_{MAE} the domain shift between the configuration trained on real events and that trained on simulation.

Setting	Train	Test	EB-NEG			EV-Eye		
			MAE↓	MPD↓	Δ_{MAE}	MAE↓	MPD↓	Δ_{MAE}
real-only	$\mathcal{D}^{\text{img}}_{\text{real}}$	$\mathcal{D}^{\text{img}}_{\text{real}}$	5.85	9.24	–	1.79	2.83	–
sim-only	$\mathcal{D}^{\text{img}}_{\text{sim}}$	$\mathcal{D}^{\text{img}}_{\text{real}}$	13.19	19.64	+7.34	55.06	87.19	+53.27
real-only	$\mathcal{D}^{\text{evt}}_{\text{real}}$	$\mathcal{D}^{\text{evt}}_{\text{real}}$	2.60	4.10	–	**3.10**	**4.60**	–
sim-only	$\mathcal{D}^{\text{evt}}_{\text{sim}}$	$\mathcal{D}^{\text{evt}}_{\text{real}}$	**2.45**	**3.77**	**-0.15**	6.55	10.76	**+3.45**

events even outperform those trained directly on real data. This behavior is likely due to the low-noise characteristics of the real event recordings, which closely match the statistics of simulated data, thereby allowing the model to exploit the cleaner and perfectly annotated synthetic samples more effectively.

4.3.2 Impact of Fine-Tuning on Real Events. We now focus our analysis exclusively on event-based data to assess the effect of fine-tuning models trained in simulation with annotated real-world events. In this configuration (*train + finetune*), a network is first trained on the synthetic domain $\mathcal{D}^{\text{evt}}_{\text{sim}}$ and subsequently fine-tuned on real annotated samples from $\mathcal{D}^{\text{evt}}_{\text{real}}$. We benchmark this approach against a (i) training solely on simulated events (*sim-only*), (ii) training solely on real events (*real-only*), and (iii) training on a mixed dataset that combines real and simulated samples within a single stage (*joint*). Since this analysis does not rely on image-domain data, we extend the evaluation to all four event-based datasets, including INI-30 and 3ET+, which provide only event-camera recordings. Quantitative results are summarized in Table 2.

Across datasets, we observe that incorporating simulated data during training consistently improves test performance compared to models trained exclusively on real data. This confirms the complementary nature of simulated and real domains: synthetic events offer perfect geometric annotations and wide variability, while real events expose the model to sensor-specific noise and contrast characteristics. Among the configurations, the three-stage approach yields the best overall results on all datasets, outperforming both the real-only and joint-domain setups. We attribute this to the structured learning process of pretraining on abundant, noise-free synthetic events followed by alignment to real sensor data, which enables the model to retain the geometric precision of simulation while adapting to real-world statistics.

Overall, these results highlight the importance of leveraging simulation in event-based eye tracking. Contrary to existing state-of-the-art methods, which rely exclusively on real recordings for training, our experiments demonstrate that incorporating simulated events consistently enhances generalization to real data across all benchmarks. The ability to pretrain on abundant, perfectly annotated synthetic samples provides a strong initialization that improves robustness even when only limited real data are available. Given the scarcity and annotation difficulty of real event-camera datasets, we argue that simulation should be regarded as a fundamental component in the training of accurate and reliable event-based gaze-tracking models.

4.3.3 Impact of Unsupervised Pretraining on Unlabeled Real Events. We continue our analysis by evaluating the effect of unsupervised pretraining on unlabeled real event data. In this setting,

Table 2. Comparison of training configurations on the Angelopoulos *et al.*, EV-Eye, INI-30 and 3ET+ datasets. We report mean absolute error (MAE, in pixels) and mean pupil distance (MPD, in pixels) for each setup. Bold values denote the best result per dataset and metric. Results obtained on a given dataset (column) are obtained on the same test set.

Method	Setting	Pre train	Train	Fine tune	EB-NEG		EV-Eye		INI-30		3ET+	
					MAE↓	MPD↓	MAE↓	MPD↓	MAE↓	MPD↓	MAE↓	MPD↓
EETNet	real-only	-	$\mathcal{D}^{\text{evt}}_{\text{real}}$	-	2.55	4.05	2.91	4.60	8.98	13.98	7.75	12.42
-	sim-only	-	$\mathcal{D}^{\text{evt}}_{\text{sim}}$	-	2.45	3.77	6.55	10.76	18.18	28.64	14.61	23.70
-	joint	-	$\mathcal{D}^{\text{evt}}_{\text{sim}} + \mathcal{D}^{\text{evt}}_{\text{real}}$	-	2.09	3.29	2.49	3.94	9.09	14.14	7.73	12.46
-	train+fine	-	$\mathcal{D}^{\text{evt}}_{\text{sim}}$	$\mathcal{D}^{\text{evt}}_{\text{real}}$	2.05	3.21	2.60	4.22	8.86	13.88	7.25	11.65
-	pretrain	$\mathcal{D}^{\text{evt}}_{\text{real}}$	$\mathcal{D}^{\text{evt}}_{\text{sim}}$	-	2.34	3.71	6.64	11.08	20.34	32.1	14.04	22.58
-	real-pre+fine	$\mathcal{D}^{\text{evt}}_{\text{real}}$	-	$\mathcal{D}^{\text{evt}}_{\text{real}}$	2.46	3.86	2.60	4.16	8.63	13.50	6.92	11.16
Ours	three-stage	$\mathcal{D}^{\text{evt}}_{\text{real}}$	$\mathcal{D}^{\text{evt}}_{\text{sim}}$	$\mathcal{D}^{\text{evt}}_{\text{real}}$	**1.96**	**3.08**	**2.36**	**3.78**	**8.37**	**13.01**	**6.76**	**10.91**

the encoder of the pupil-tracking network is first pretrained using the autoencoder described in Section 3.2.2, and the resulting weights are then used to initialize the supervised network before training on synthetic events. We evaluate three configurations: (i) pretraining on real events followed by supervised training on simulated events (*pretrain*), (ii) pretraining on real events followed by finetuning on real data (*real-pre+fine*) and (iii) the same configuration as (i) where we additionally finetune on annotated real data (*three-stage*). These are compared to their non-pretrained counterparts discussed in the previous section.

For this analysis, the unlabeled real data $\mathcal{D}^{\text{evt}}_{\text{real,u}}$ was selected according to the annotation protocols specific to each dataset. For the Angelopoulos et al.'s EB-NEG dataset, we used the unlabeled user recordings and manually defined a 157×90 pixel region of interest centered on the eye. For EV-Eye, we exploited the four-second intervals between consecutive labels, defining the region of interest around the centroid of the annotations. For INI-30, we used the midpoint between annotated samples and downsampled them following the procedure described in Section 4.2.2. For all datasets, events were accumulated into frames using a 5 ms integration window. Data from the test sets were excluded from this process to avoid contamination. An example of the autoencoder's reconstruction is shown in Figure 2, illustrating its ability to recover the spatial structure of the eye region and the polarity distribution of real events.

Results are summarized in Table 2. Overall, unsupervised pretraining on real events improves performance across most configurations, confirming that adapting the encoder to real sensor statistics before supervised training enhances generalization to the target domain. When focusing on models trained solely on real-world data, and comparing them with the ones trained on simulated events, adding pretraining consistently leads to performance gains, with the exception of the EV-Eye dataset, where a slight degradation is observed. We attribute this to the higher noise levels and variability in EV-Eye's real recordings, which may cause the encoder to learn feature representations biased toward sensor noise and artifacts, limiting its ability to effectively exploit simulated data during subsequent supervised training. This limitation is effectively mitigated in the proposed three-stage training procedure, where unsupervised pretraining on unlabeled real events is followed by supervised training on simulated data and final fine-tuning on labeled real recordings, achieves the most consistent and highest performance across all datasets, outperforming real-only trained models by up to 25%, real-pretrained and fine-tuned models by up to 20%, and simulation-pretrained models fine-tuned on real data by up to 10%. These results clearly indicate

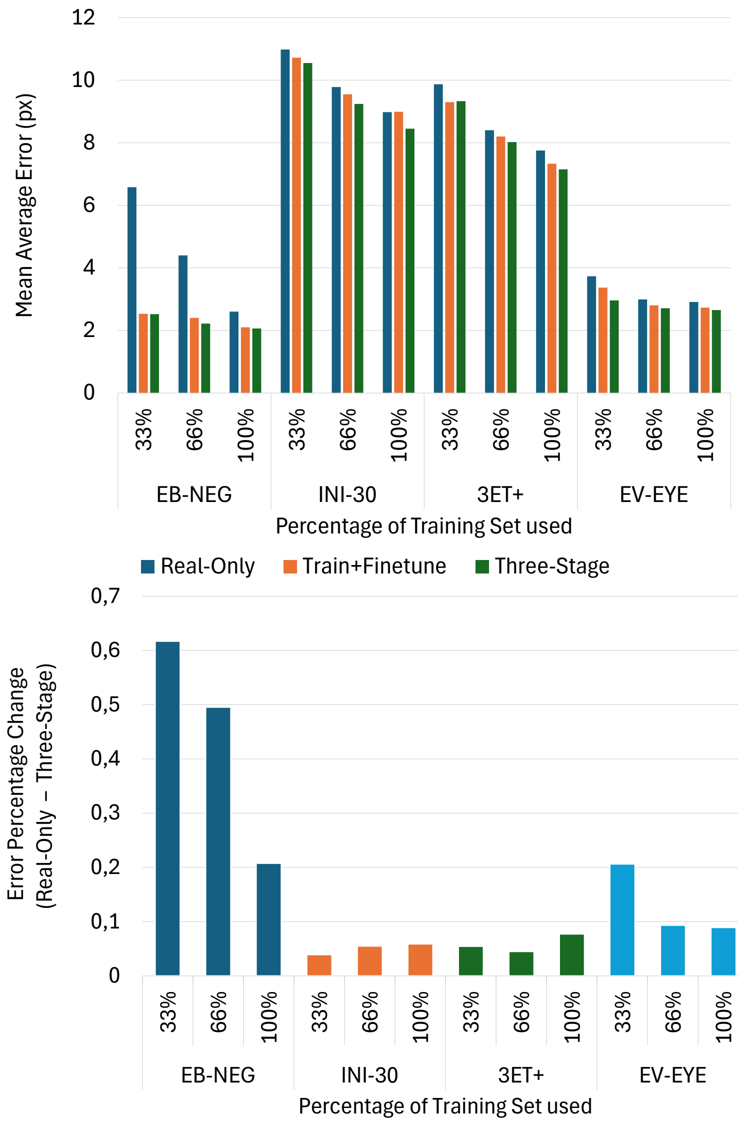


Fig. 4. Comparison of method performance with varying training set size. (Top) Mean Absolute Error (MAE) across the three training procedures. (Bottom) Percentage MAE difference between the Real-Only and Three-Stage versions

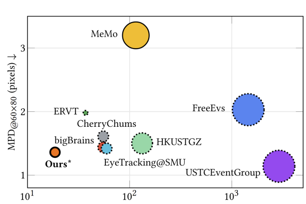
that the combination of unsupervised adaptation to real event statistics and progressive supervision across simulated and real domains yields the most robust and accurate event-based eye tracker.

4.3.4 Training set reduction Ablation study. We analyzed performance variation with respect to training set size by retraining the *real-only*, *train+finetune*, and *three-stage* models using 33%, 66%, and 100% of the available $\mathcal{D}_{\text{real}}^{\text{evt}}$ training data. As shown in Figure 4, all datasets benefit from our

Table 3. Comparison of our method with baseline approaches on INI-30 and 3ET+ following the official evaluation strategy, i.e., with models evaluated at a lower resolution space (64×64 and 60 × 80 respectively). We mark with * the performance obtained predicting outputs in the original resolution (as in Tables 1-2) and then re-mapping them at the benchmark scale (×1/8). With $ we indicate MACs estimated from FLOPs (as MACs ≈ 0.5·FLOPs). Gray rows indicate methods exploiting temporal reasoning, which are thus not directly comparable with our method. We report the mean pupil distance (MPD, in pixels). Best results among comparable methods are shown in **bold**, and second-best results are underlined. The plot provides a visual comparison of the 3ET+ results, where methods employing temporal processing are shown with dotted contours, and non-temporal methods with solid ones. Dot radius is proportional to the number of model parameters.

INI-30 — $^{1}\text{MPD}_{@64\times64}$

Method	Pred.	MACs	MPD^1 ↓
Retina [Bonazzi et al.]	64×64	3.03M	3.24
GG-SSM [Zubic at al.]	64×64	3.01M	3.11
EETNet [Aspesi et al.]	64×64	23.97M	2.77
Ours	64×64	23.97M	2.48
Ours*	640×480	**18.60M**	**1.63**



3ET+ — $^{2}\text{MPD}_{@60\times80}$

Method	Pred.	Temporal Proc.	MACs	MPD^2 ↓
3ET+ 2024 Challenge [Wang et al. 2024a]:				
bigBrains	60×80	Temp.-CNN	55.2M	1.44
USTCEventGroup	60×80	MambaPupil	-	1.67
ERVT	60×80	ConvLSTM	37M	1.98
FreeEvs	60×80	GRU	1.45G$	2.03
Go Sparse	60×80	GRU	-	3.51
EFFICIENT	60×80	Attention + BiLSTM	-	3.51
GTechVision	60×80	ConvLSTM	-	4.94
MeMo	60×80	-	115M$	3.20
3ET+ 2025 Challenge [Chen et al. 2025]:				
USTCEventGroup	60×80	Temp.-Attention	2.9G	1.14
EyeTracking@SMU	60×80	Temp.-Filter & Flow	59.54M	1.42
HKUSTGZ	60×80	GRU	132.5M$	1.5
CherryChums	60×80	Temp.-CNN	55.18M	1.61
EETNet [Aspesi et al.]	60×80	-	28.99M	3.91
Ours	60×80	-	28.99M	3.24
Ours*	640×480	-	**18.60M**	**1.36**

method, with the largest improvements observed for EB-NEG and EV-Eye. We attribute this to the fact that these two datasets contain fewer labeled samples but a large amount of unlabeled data, which our approach effectively leverages. Additionally, improved simulation quality may contribute to these gains. In these cases, the plot on the right illustrates how utilizing unlabeled data compensates for reduced labeled data. Conversely, in the other datasets, while pretraining still improves performance, it is less pronounced due to the absence of abundant unlabeled data with visual features different from those used for training, as they originate from the same users.

4.3.5 State of the Art Comparison. We conclude our experimental validation by comparing our approach with state-of-the-art models on the INI-30 and 3ET+ datasets. Unlike the previous experiments, where the network was given a downsampled frame but still required to predict for valuation at the original resolution, we follow here the official evaluation protocols of INI-30 and 3ET+. In both, networks are evaluated in the downsampled resolution (64 × 64 and 60 × 80, respectively). For completeness, we also report results from our model when predicting at the original resolution (denoted as *Ours**) and subsequently rescaled to the benchmark evaluation scale, as we found our network to benefit from predicting original scale pupil centers. Results show that our approach achieves state-of-the-art performance without relying on temporal filtering, attention mechanisms, or recurrent architectures, while maintaining a compact model size suitable for deployment on embedded accelerators.

Table 4. The **final position** denotes the configuration used in the **sim-only** experiments reported in Table 2.

Method	MAE	MPD
Initial Test	2.73px	4.32px
Wrong zoom	2.77px	4.42px
Final Position	2.45px	3.87px

4.4 Limitations

The proposed method focuses on continuous pupil tracking under smooth eye motion and does not explicitly model eye blinks or strong eyelid occlusions. This design choice follows the assumptions adopted during simulation and training, where pupil visibility is maintained, and may lead to degraded performance in scenarios with frequent blinks or prolonged occlusions. Incorporating explicit blink detection or temporal state modeling is a natural direction for future work. Due to the lack of available datasets and simulation setups, this work does not evaluate robustness to unseen population characteristics. Instead, it demonstrates cross-subject generalization through user-based train, validation, and test splits. Moreover, this method still relies on manual adjustment of the simulated camera position.

To study the impact of camera pose on sim-to-real transfer, we trained models using synthetic events generated with different virtual camera configurations and evaluated them on real Angelopoulos et al. data. Table 4 summarizes these preliminary results, highlighting the sensitivity of performance to camera pose alignment.

5 Conclusion

We presented a unified training framework for event-based pupil tracking that achieves robust real-world performance by combining realistic simulation with both labeled and unlabeled real data through supervised and unsupervised learning stages. To enable this, we extended the U2Eyes renderer with the v2e event camera simulator, generating realistic event streams under controlled motion and illumination. Experiments on four public benchmarks demonstrate that models trained solely on simulated events generalize effectively to real data, exhibiting a smaller domain gap than image-based counterparts and, in some cases, outperforming models trained directly on real recordings. Building on this observation, we proposed a progressive three-stage training strategy that integrates unsupervised pretraining on unlabeled real events, supervised learning on synthetic data, and fine-tuning on annotated real recordings. This approach consistently achieved the best results across all datasets, improving pupil localization accuracy by up to 25% over real-data-only baselines. Our findings show that events, being less sensitive to texture and illumination, are easier to simulate faithfully and transfer more robustly to real near-eye event streams. Overall, this work demonstrates that combining simulation and self-supervised learning can effectively overcome data scarcity in event-based vision, enabling robust pupil tracking for future low-power wearable systems.

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